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External validation of a deep learning detection system for glaucomatous optic neuropathy: a real-world multicentre study

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OBJECTIVES: To conduct an external validation of an automated artificial intelligence (AI) diagnostic system using fundus photographs from a real-life multicentre cohort.

METHODS: We designed external validation in multiple scenarios, consisting of 3049 images from Qilu Hospital of Shandong University in China (QHSDU, validation dataset 1), 7495 images from three other hospitals in China (validation dataset 2), and 516 images from high myopia (HM) population of QHSDU (validation dataset 3). The corresponding sensitivity, specificity and accuracy of this AI diagnostic system to identify glaucomatous optic neuropathy (GON) were calculated.

RESULTS: In validation datasets 1 and 2, the algorithm yielded accuracy of 93.18% and 91.40%, area under the receiver operating curves (AUC) of 95.17% and 96.64%, and significantly higher sensitivity of 91.75% and 91.41%, respectively, compared to manual graders. On the subsets complicated with retinal comorbidities, such as diabetic retinopathy or age-related macular degeneration, in validation datasets 1 and 2, the algorithm achieved accuracy of 87.54% and 93.81%, and AUC of 97.02% and 97.46%, respectively. In validation dataset 3, the algorithm achieved comparable accuracy of 81.98% and AUC of 87.49%, with a sensitivity of 83.61% and specificity of 81.76% on GON recognition specifically in the HM population.

CONCLUSIONS: With acceptable generalization capability across varying levels of image quality, different clinical centres, or certain retinal comorbidities, such as HM, the automatic AI diagnostic system had the potential to provide expert-level glaucoma detection.

Eye (2023) 37:3813–3818; https://doi.org/10.1038/s41433-023-02622-9

INTRODUCTION

Glaucoma, a progressive optic neuropathy characterized by atrophy of optic nerve and visual field defects, significantly affects both public health and quality of life of affected individuals [1]. By 2040, an estimated 111.8 million people worldwide will be affected by glaucoma, with approximately 66.8 million of those cases occurring in Asia [2, 3]. In China, the prevalence of glaucoma in people over 40 years old is about 2.6%, with an increasing incidence with age [4]. According to reports, over 90% of glaucoma cases in China may go undiagnosed, with a blindness rate as high as 30% [5, 6]. Since intraocular pressure reduction is currently the only clinical treatment available to reduce the risk of vision loss, accurate diagnosis and adequate treatment are crucial for disease management [7]. Therefore, objective and automated methods are urgently needed for proactive screening of glaucoma in China and other regions with a high incidence of the disease.

Recently, artificial intelligence (AI) techniques have emerged as a promising approach for automated glaucomatous optic neuropathy (GON) diagnosis, utilizing deep learning (DL) models to extract discriminative features from raw fundus images and make accurate predictions [8]. Recent studies have presented DL-based models based on the cup-to-disc ratio (CDR) for GON detection, with area under the receiver operating curves (AUC) of 0.986 and 0.996, surpassing traditional methods [9, 10]. In another study, researchers proposed a machine-to-machine (M2M) DL algorithm trained on spectral domain optical coherence tomography (SD-OCT) to identify glaucomatous damage in fundus photographs, achieving an AUC of 0.801 and a mean absolute error of 7.39 µm for the retinal nerve fibre layer (RNFL) thickness estimation [11]. Furthermore, ensemble model, that combines imaging with demographic and clinical features to make predictions, has been investigated, yielding higher accuracy than single models [12]. However, the present studies fail to provide any information on the performance of these models when executing a multitude of high-fidelity simulations.

Even though numerous studies have demonstrated the accuracy of glaucoma detection algorithms, their coverage or performance has lagged behind that of ophthalmologists. The algorithms that are currently being utilized for deep learning detection systems for GON are laden with a myriad of limitations. Since fundus images' quality varied greatly among various fundus cameras and imaging methodologies, existing algorithms for GON

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Received: 23 November 2022 Revised: 17 May 2023 Accepted: 2 June 2023
Published online: 15 June 2023

Table 1. Baseline characteristics of participants in Training and Validation Datasets.

Datasets	Source	Total	Gender		Age (y)	Glaucoma	Normal	Comorbidities
			Male	Female				
Training dataset	Chinese Glaucoma Study Alliance	20,877	10,806 (51.76)	10,071 (48.24)	52.49 (20.72)	6385 (30.58)	10,647 (51.00)	3845 (18.42)
Validation dataset 1	QHSDU	3049	1636 (53.66)	1413 (46.34)	47.29 (22.93)	400 (13.12)	2427 (79.60)	324 (10.63)
Validation dataset 2	Three other hospitals	7495	3517 (46.92)	3978 (53.08)	53.75 (16.83)	547 (7.30)	4434 (59.16)	2617 (34.92)
Validation dataset 3	QHSDU	516	243 (47.09)	273 (52.91)	45.82 (15.63)	61 (11.8)	N/A	455 (88.2)

Continuous variables are presented by mean (standard deviation) and categorical variables are presented by *N* (%).

identification developed utilizing fundus images from a single centre were unable to provide comfortable classification performance throughout complicated databases [13]. Moreover, the algorithms may not function well in real-world scenarios because the datasets used to train and evaluate them were frequently neither normal nor glaucoma-affected eyes [14, 15]. Multiple comorbidities such as diabetic retinopathy (DR), age-related macular degeneration (AMD), and especially high myopia (HM) are the most common causes of high false positive detection rates. All these challenges limit the clinical application of AI diagnosis on glaucoma and delay its integration into the health care systems. External validation is necessary to determine a model's reproducibility and generalizability to new and different patients [16]. In order to truly realize the transformation from academic research to clinical practice, it is essential to evaluate the generalization capability and robustness of the models using real-world multicentre datasets.

Previously, our group developed an automatic screening system for identifying GON by directly analysing retinal fundus images [17]. To ensure the accuracy of the GON classification model in this study, we implemented an image quality control module to automatically remove low-quality photos during the image acquisition process. Then, for the segmentation and categorization of GON with good generalization performance across multiple clinics or several concurrent retinal disorders, notably HM, a fully automatic robust glaucoma screening system was established. The multi-task convolution neural network model based on ResNet, with the main task of glaucoma classification and the auxiliary task of optic disc atrophy, exhibited a sensitivity of 95.9%, 95.4%, and 95.7%, respectively, and a specificity of 97.7%, 91.6%, and 92.2%, respectively. Moreover, the AUC was 0.993, 0.968, and 0.974, respectively, indicating excellent performance in the training set and validation set. This study's objectives were to confirm the system's efficacy and efficiency and demonstrate the utility of its applications in clinically relevant, multicentre real-world datasets. The iterative external validation process serves as a valuable tool for providing constructive feedback to guide previous model improvement efforts, ultimately enhancing the performance and applicability of the deep learning model. This procedure acts as a safeguard against overfitting or other forms of bias in the model results, ensuring the model's reliability and robustness. The repetitive validation procedure reinforces the model's generalizability, making it more adaptable to various real-world settings and populations. The aforementioned benefits make the external validation process an essential component in the development and refinement of deep learning models.

MATERIALS AND METHODS

Ethical statement and datasets

In order to validate the effectiveness, generalizability, and robustness of the automatic GON diagnosis system in cross-dataset evaluation, the performance was evaluated in a real-world cohort from hospitals in various locations of China, including one hospital from North China (i.e., Qilu Hospital of Shandong University (QHSDU)), and three other hospitals in South China (i.e., Guangdong General Hospital, Guangdong Shuidian Hospital and Nanning No.1 People's Hospital). To be specific, validation dataset 1 included 3049 images from QHSDU, while validation dataset 2 contained 7495 images from the other three hospitals. The images in these two datasets, which were collected from the general public, may exhibit concurrent diseases such as DR or AMD. To construct the validation set, one fundus image was randomly selected for each patient. The diagnosis of DR and AMD for each patient was established by expert ophthalmologists specializing in retinal diseases. The diagnosis was based on the patient's medical history and ocular examination results. In cases where necessary, additional diagnostic tools, such as optical coherence tomography (OCT) or fundus fluorescein angiography, were employed. To assess the GON classification performance for the HM population, validation dataset 3 was created using the 516 images of HM patients from the QHSDU, with an average axial length of 25.51 ± 1.53 mm. Details of all the datasets were described in Table 1.

This study was approved by our institutional review boards (IRB) and performed following the relevant guidelines and regulations, including the Declaration of Helsinki. Patients' informed consents were exempted by IRB of each source, because of the fully anonymized retrospective usage of fundus images in this study.

Algorithm development

In this study, the proposed deep learning system (DLS) for automatic GON diagnosis was developed by Tencent Healthcare (Shenzhen, China) [17]. Supplementary Fig. 1 depicted the workflow of the deep learning system for automatic GON diagnosis. A Quality Assessment (QA) module initially evaluated the image quality of the original colour fundus images. Images that satisfied the quality criteria and were approved by the QA module were fed into the GON model, which categorized the images as referable or non-referable GON. Based on the AI prediction result, a report included general information of patients, summary of diagnostic information, and further recommendations would be generated for the reference of clinicians and patients. The whole procedure could be performed either online or offline, providing various deployment options for diverse practical scenarios.

Quality assessment model development. The variation of image quality is a major factor that negatively affects the performance of the DLS when being adopted in a real-world clinical scenario. Therefore, when developing the automatic GON diagnosis system, the importance of the QA module was taken into consideration. Residual Neural Network (ResNet 18), a well-established and effective convolutional network architecture, was used to categorize the quality of colour fundus images into three categories: good quality, acceptable quality, and poor quality. The good quality colour fundus images referred to images that could observe the cup, disc, and vessel structures, without blur, or over- or under-exposure problems. The term "poor-quality images" referred to images that made it difficult to reasonably diagnose GON. It varied a lot, but generally contained images with severe blur, moving or out of focus; severe under/over exposure; images without optic disc; black and white fundus images, etc. Images that lay between good quality and poor quality to some extents were classified as acceptable quality. The quality assessment results and the classification performance with and without poor-quality images were illustrated in Supplementary Table 1 and Table 2. Representative examples of images with different qualities were listed in Supplementary Fig. 2.

Images were resized to 512 × 512 pixels in order to maintain the fine details while not consuming excessive processing resources and pre-processed to rescale the intensity between 0 and 1 before being fed to the network. Data augmentations of random rotation, cropping, horizontal flipping, and colour jitting were used in the training procedure to expand the data variations.

Table 2. Criteria for GON Grading.

Classification	Pathologic features
Non-GON	None of the GON appearance
Suspected GON	At least 2 conditions positive: 1. 0.7 ≤ Vertical cup-to-disc Ratio ≤ 0.85; 2. Rim width ≤0.1-disc diameter; 3. General rim thinning ≥60° or localized rim thinning <60° in the superior region or inferior region; 4. Retinal nerve fiber layer defects; 5. Splinter hemorrhages on disc; 6. β peripapillary atrophy.
Confirmed GON	Any of the following conditions: 1. Vertical cup-to-disc Ratio ≥ 0.85; 2. Retinal nerve fiber layer defects correspond with the thinning area of rim or notches.

GON glaucomatous optic neuropathy.

GON classification model development. The classification of GON was performed on the disc and surrounding region rather than the entire image in order to incorporate the clinical domain knowledge into the DLS. The GON model framework was established in a two-stage manner, as shown in Supplementary Fig. 3. A pre-trained U-Net segmentation network was used in the first stage to segment the optic disc. Three Disc Diameter (3DD) region surrounding the disc was then cropped, set as the region-of-interest (ROI), and provided to the GON classification model in the second stage to determine if the image is likely to be GON or not.

ResNet-34 was adopted as the framework for the classification network according to the criteria of GON defined in Table 2 [18]. The cropped ROI images were resized to the dimension of 256 × 256 and rescaled to the range of 0–1, before being fed to the GON classification network. The vertical CDR and RNFL defects were considered the major structural image cues for glaucoma detection [19].

Additionally, fundus images with other diseases, e.g., DR, AMD, pathological myopia, etc, were included in both referable GON and non-referable GON classes to distinguish GON from other diseases and improve the generalization capability of the deep learning system.

To enhance our ability to distinguish GON from myopic optic atrophy, we also implemented a multi-task deep learning framework that simultaneously predicted two tasks associated with the optic disc region (glaucoma and optic disc atrophy).

For validation datasets 1 and 2, images were graded via a two-round adjudication process with two certified ophthalmologists and one senior glaucoma specialist involved. The reference standard was built up when the two graders independently assessed referable GON diagnosis using fundus images alone and reached a consensus on the images in the first round; the discordant images were further adjudicated by the senior glaucoma specialist in the second round. For validation dataset 3, one glaucoma specialist accessed the glaucoma diagnosis with a full glaucoma workup, including the medical history of patients, IOP measurement, fundus images, visual field (VF) test, and OCT scans. After this process, images were categorized as Non-GON, suspected GON, and confirmed GON. Referable GON was defined as suspected GON or confirmed GON.

Visualization of referable GON prediction

Beyond the GON prediction, a heatmap image was generated by the network model to highlight the GON-related pathological regions on the input image. The heatmap visualization (Supplementary Fig. 4) identified the 3DD region surrounding the optic disc for the pathological features of AMD, DR, and myopia eyes. On the basis of the visualization output, the clinical practitioners could substantiate the validity of the DL system and promote the clinical adoption of the automated GON diagnostic system, especially for those underserved populations.

Comparison of DLS versus general ophthalmologists

Additionally, two general ophthalmologists with more than ten years of relevant work experience were employed to compare the diagnostic performance of the GON classification model. They were not involved in the creation of the reference standard and were blind to the ground truth. The two ophthalmologists were asked to classify the fundus images from validation datasets 1 and 2 into GON and non-GON.

Statistical analysis

To evaluate the performance and discriminatory ability of the developed automated GON diagnosis system, the area under the receiver operating characteristic curve (AUC) was calculated with a 95% confidence interval (CI). Besides, sensitivity and specificity were also calculated using Stata software, version 14.0 (Stata Corp., College Station, TX).

RESULTS

Evaluation of the efficiency and generalization of detecting GON on databases from a real-world clinical scenario

We first evaluated the efficiency of the automatic GON diagnosis system on validation datasets 1 and 2, in which referable GON was assessed using the fundus images alone. The algorithm achieved AUC of 0.952 (95% CI: 0.945–0.968) and 0.966 (95% CI: 0.959–0.974), respectively. The algorithm kept significantly higher and consistent sensitivity (0.918 and 0.914, respectively) in both datasets. Comparatively, the sensitivity of the two manual graders was 0.788 and 0.928 for dataset 1, and 0.603 and 0.794 for dataset 2, which served as the validation dataset. On both datasets, the algorithm achieved good specificity (0.934 and 0.922, respectively), although slightly lower than manual scores (0.968 and 0.983, respectively). Detailed validation results as well as the receiver's operating characteristic curves were listed in Table 3 and Fig. 1.

In addition, we evaluated the generalization capability of the automatic GON diagnosis system on images with/without the co-existence of some retinal diseases such as DR and AMD. The algorithm demonstrated excellent discrimination capability on GON with these retinal comorbidities in both datasets, with AUC of 0.970 (95% CI: 0.954–0.987) and 0.975 (95% CI: 0.964–0.985), respectively. Meanwhile, the algorithm revealed good performance with high AUC on diagnosing referable GON on fundus images without the comorbidities as well, as shown in Table 4. In terms of sensitivity and specificity, the algorithm achieved relatively good performance on both subsets with/without these comorbidities.

Evaluation of the glaucoma classification performance on fundus images of high myopia population

We further evaluated the performance of the automatic GON diagnosis system on images of the HM population in validation dataset 3. As the HM will negatively affect the disc and surrounding regions and make it difficult to detect the signs of glaucoma based solely on fundus images, diagnosing glaucoma on HM fundus images proved particularly tough for AI models [8, 20]. Note that the gold standard for the GON label on validation dataset 3 was drawn based on a full glaucoma workup, including the medical history of patients, IOP measurement, fundus images, VF tests, and OCT scans. For the images of the HM population, the performance of the algorithm dropped dramatically compared with that of validation dataset 1 and validation dataset 2, but still kept relatively acceptable, with an AUC of 0.875 (0.816–0.934), a sensitivity of 0.836 and a specificity of 0.818 (Supplementary Table 3).

DISCUSSION

In this study, to cope with the challenges of existing work, we developed a DLS for categorizing referable GON based on the pathological characteristics visible on fundus pictures in the region around the optic disc. Images from the training dataset were either normal or had specific retinal concomitant conditions like HM, DR, or AMD, reducing the possibility of bias and increasing real-world clinical relevance. Furthermore, a real-world cohort from different hospitals in various regions of China was used to test the performance of the algorithm. Although the number of studies reporting GON detecting systems has been

Table 3. Classification performances of the system and the ophthalmologists on validation datasets 1 and 2.

	Sensitivity (95% CI)		Specificity (95% CI)		AUC (95% CI)		*P*-value	
	Set 1	Set 2	Set 1	Set 2	Set 1	Set 2	Set 1	Set 2
Grader 1	0.788 (0.773–0.802)	0.794 (0.785–0.803)	0.975 (0.969–0.980)	0.983 (0.980–0.986)	N/A	N/A	8.43e–9	9.04e–39
Grader 2	0.928 (0.918–0.937)	0.603 (0.592–0.614)	0.977 (0.971–0.982)	0.968 (0.983–0.988)	N/A	N/A	1.18e–4	9.04e–39
AI Model	0.918 (0.908–0.927)	0.914 (0.907–0.920)	0.934 (0.925–0.942)	0.922 (0.916–0.928)	0.952 (0.945–0.968)	0.966 (0.959–0.974)		

AUC area under the curve, *CI* confidence interval, *AI* artificial intelligence, Set 1: validation dataset 1, Set 2: validation dataset 2, *N/A* not available.

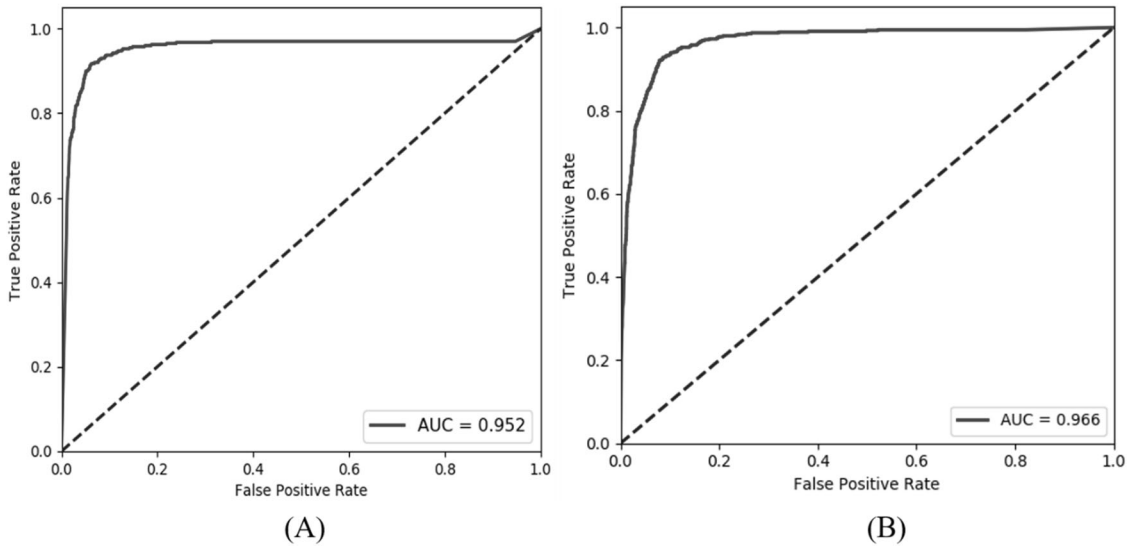


Fig. 1 The receiver operating characteristic analysis of GON classification performance in validation datasets. A The area under the curve (AUC) validation for dataset 1 is 0.952. **B** AUC for validation dataset 2 is 0.966.

Table 4. Evaluation of the generalization capability of the algorithm on images with/without the features of some retinal comorbidities such as DR and AMD on validation dataset 1 and validation dataset 2.

	Sensitivity (95% CI)		Specificity (95% CI)		AUC (95% CI)		*P*-value	
	Set 1	Set 2	Set 1	Set 2	Set 1	Set 2	Set 1	Set 2
With Comorbidities	1.000 (1.000–1.000)	0.922 (0.916–0.928)	0.825 (0.811–0.838)	0.939 (0.933–0.944)	0.970 (0.954–0.987)	0.975 (0.964–0.985)	4.55e–11	1.97e–6
Without Comorbidities	0.893 (0.882–0.903)	0.912 (0.905–0.918)	0.944 (0.935–0.952)	0.913 (0.906–0.919)	0.943 (0.922–0.965)	0.963 (0.954–0.972)		
Overall Performance	0.918 (0.908–0.927)	0.914 (0.907–0.920)	0.934 (0.925–0.942)	0.922 (0.916–0.928)	0.952 (0.945–0.968)	0.966 (0.959–0.974)		

DR diabetic retinopathy, *AMD* age-related macular degeneration, *AUC* area under the curve, *CI* confidence interval, Set 1: validation dataset 1, Set 2: validation dataset 2.

rapidly increasing, too much effort was being focused on developing new algorithms rather than externally validating and assessing the impacts of ones that already exist. The DLS showed excellent diagnostic performance, generalization, and robustness in the multicentre real-world datasets, suggesting promising clinical usage with a higher evidence level. We optimized the previously studied DLS to enhance its convenience, reliability, and accuracy for application in real-world medical settings. Specifically, our study simulated scenarios present in primary care hospitals and eye screening programs where only fundus photography is available, no other eye examination equipment is accessible, and patients have complex profiles. The outcomes of the study demonstrated that the model can accomplish robust detection performance even with unprocessed data, providing greater value than existing model applications.

Our automatic GON diagnostic system exhibits a robust performance on the prediction of referable GON with access to only a single fundus image. An AUC of 0.952 was achieved on validation dataset 1, and an AUC of 0.966 on validation dataset 2. In validating datasets 1 and 2, the algorithm has more consistent sensitivity and specificity than human scorers. However, the specificity of human graders was slightly higher than that of the algorithm in both validation datasets, indicating that the algorithm needs further refinement in its specificity in the future. Currently, higher sensitivity on glaucoma detection might bring more benefit for the disease screening in primary health care settings to avoid missing potential glaucoma patients.

As we know, high-quality images were often the first step in developing successful models for GON diagnostic applications [9, 18, 21]. Real-world fundus images from hospitals or clinics generally suffered from various levels of image qualities, which might negatively affect the performance of AI models. In practice, a continuous trickle of low-quality photos might indicate that the patient needs to be referred to an ophthalmologist for additional ocular examinations. To address this problem, we installed a quality control module in the local computer linked to the fundus camera, which could give the operator immediate feedback on whether the captured images meet the minimum standards for image quality and lessen the possibility that poor image quality would then influence the algorithm's performance.

The influence of various retinal comorbidities, such as HM, DR, or AMD, which were frequently the causes of false negative or false positive glaucoma predictions, was another significant element in the generalization of DLSs [22]. In order to alleviate the influence of these diseases, we specifically added the fundus images of these diseases as a negative control in the training database. In addition, the model specifically focused on the 3-disc diameter region surrounding the optic disc, avoiding missing the major structural image cues directed to glaucoma diagnosis, including vertical CDR of more than 0.7, rim thinning or notching, disc haemorrhages, RNFL defect, or β PPA, etc. [9]. Meanwhile, the region of interest in the disc region safely excluded the influence of DR or AMD, the pathologies of which are mainly distributed on the whole retina or located in the macula region. To validate our DL model's general applicability, we obtained fundus images from different hospitals in China. For images with the aforementioned retinal comorbidities, the system performed superbly on referable GON detection, with accuracy of 87.54% and 93.81%, and the AUC of 97.02% and 97.46%, respectively, in Validation sets 1 and 2. The automated GON diagnosis system showed favourable adoption in real-world settings.

The method also showed satisfactory performance on HM fundus images in validation dataset 3 with an accuracy of 81.98% and AUC of 87.49% (sensitivity, 83.61%; specificity, 81.76%), respectively. HM has been considered a major risk factor for glaucoma, and numerous population-based pieces of research have thoroughly investigated the strong relationship between the two diseases [23, 24]. Based on findings from a meta-analysis, the association between the degree of myopia and the prevalence of glaucoma increases dose-dependently, especially in HM [25]. Persons with HM are at a 7.3-fold greater risk of glaucoma than those without myopia in a 10-year follow-up [26]. However, for a combination of reasons, it might be challenging to apart glaucoma from HM based on manual morphological assessment, not only for AI models but for human experts as well, when screening GON based solely on fundus images [27, 28]. The structural characteristics of HM on the disc region, such as increased pallor in the rim, shallow cups, tilting, torsion, and peripapillary atrophy, make it challenging for human graders to identify myopic glaucoma [29]. In addition, traditional methods like optical coherence tomography (OCT) and visual field are not able to correctly differentiate glaucomatous lesions because of the functionally abnormal caused by macular atrophy and RNFL thickness decreases with increasing axial length in non-glaucomatous myopic eyes. Methodology to differentiate the structural change caused by glaucoma from that caused by myopia on the optic nerve calls for a wider and more thorough investigation, given the rising incidence of both myopia and glaucoma, especially in East Asia [30]. This study developed a multi-task deep learning model based on the area surrounding the optic disc, which used the optic disc and its related regions as the primary task, and the features of other diseases in the optic disc region as the secondary task. Compared with single-task models, multi-task models can be useful for differential diagnosis, especially of HM and glaucoma. Our well-established DL model was effective in accurately identifying glaucomatous damage from HM eyes on fundus images in the real-world scenario, which could assist in improving the clinical decision-making process.

Despite the good performance of our novel DL system, there were still several limitations. Firstly, with regards to the validation procedure, while we employed three distinct datasets from two diverse sources to validate our results, the limited quantity of data and the absence of external validation in additional populations limit the generalizability of our findings. Secondly, our algorithmic approach was founded on a deep learning model that was trained on labelled datasets. The labelling process itself may introduce some degree of subjectivity or errors, which may impact the model's accuracy. Integrated with other clinical parameters, including IOP measurements, VF test, and OCT, DL systems would improve the sensitivity to recognize GON on fundus images. Glaucoma informativity biomarkers might provide a precise diagnosis in the future [31]. The accuracy and AUC of the algorithm still needed to be improved, when there were several retinal concomitant disorders present, notably in the HM group. In addition, varied anatomical and clinical characteristics as well as different incidences of glaucoma were related to ethnicity [1, 32]. All of the data was taken from the Chinese population, hence more confirmation in other populations was definitely required in the future.

Given a large number of glaucoma patients in China, the priority task of AI diagnostic systems was to apply its real-world deployment to the health care system, especially in resource-limited areas, to discover glaucoma in asymptomatic individuals and expand to population-level screening systems. This was especially important in situations with limited resources. The proposed automated GON diagnosis system achieved excellent performance and generalization capability for the detection of referral GON even complicated with multiple retinal diseases, especially HM. Therefore, this system could provide potentially efficient application in the primary care settings or health care systems for large-scale screening of glaucoma in undiagnosed patients and monitoring glaucoma progress, providing the opportunity to intervene before permanent vision loss occurs.

SUMMARY

What was known before

- Artificial intelligent techniques are employed to facilitate the detection of GON on high quality fundus images.
- The coverage or performance of existing algorithms for the detection of GON has lagged behind that of ophthalmologists.
- It is essential to evaluate the generalization capability and robustness of the models using real-world multicentre datasets.

What this study adds

- The proposed automated GON diagnosis system achieved excellent performance and generalization capability for the detection of referral GON even complicated with multiple retinal diseases, especially HM.
- It could provide potentially efficient application for large scale screening on glaucoma in undiagnosed patients and monitoring glaucoma progress.

DATA AVAILABILITY

Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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ACKNOWLEDGEMENTS

The authors express their sincere gratitude to the patients who participated in the trial.

AUTHOR CONTRIBUTIONS

QY was responsible for the concept and design. All of the authors were responsible for the acquisition, analysis, or interpretation of data. XQ and SX wrote the original manuscript draft. All authors contributed to the critical revision of the manuscript. XQ and SY were responsible for statistical analysis. QY supervised the project. QY had full access to all of the data in the study and took responsibility for the integrity of the data and the accuracy of the data analysis.

COMPETING INTERESTS

The authors declare no coompeting interests.

ADDITIONAL INFORMATION

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41433-023-02622-9.

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